7.6 Inverse Trigonometric Functions

The six basic trigonometric functions are not 1-1. So to have sensible meaning for their inverses, we can restrict their domains to intervals on which they are 1-1.

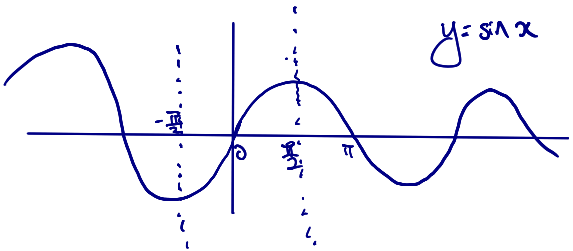
For example, by restricting the domain to $[-\frac{\pi}{2}, \frac{\pi}{2}]$, the sine function is 1-1, and hence has an inverse.

Under suitable restricted domains, these trigonometric functions are 1-1, and they have inverses, which we can denote as follows.

$$y = \sin^{-1} x \text{ or } y = \arcsin x \qquad y = \cos^{-1} x \text{ or } y = \arccos x$$

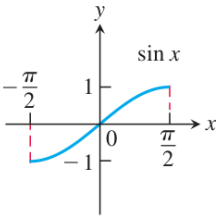
$$y = \tan^{-1} x \text{ or } y = \arctan x \qquad y = \cot^{-1} x \text{ or } y = \operatorname{arccot} x$$

$$y = \sec^{-1} x \text{ or } y = \operatorname{arcsec} x \qquad y = \csc^{-1} x \text{ or } y = \operatorname{arccsc} x$$

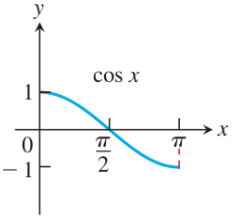


The graphs of trigonometric functions under suitable restricted domains.

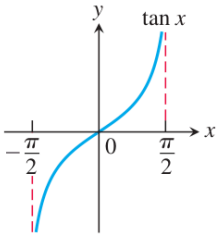
Domain restrictions that make the trigonometric functions one-to-one



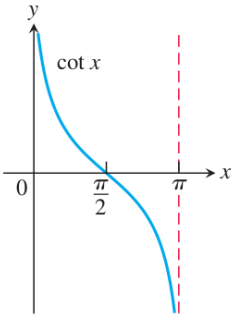
$y = \sin x$
Domain: $[-\pi/2, \pi/2]$
Range: $[-1, 1]$



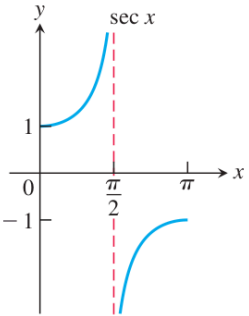
$y = \cos x$
Domain: $[0, \pi]$
Range: $[-1, 1]$



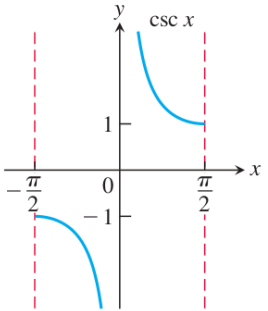
$y = \tan x$
Domain: $(-\pi/2, \pi/2)$
Range: $(-\infty, \infty)$



$y = \cot x$
Domain: $(0, \pi)$
Range: $(-\infty, \infty)$

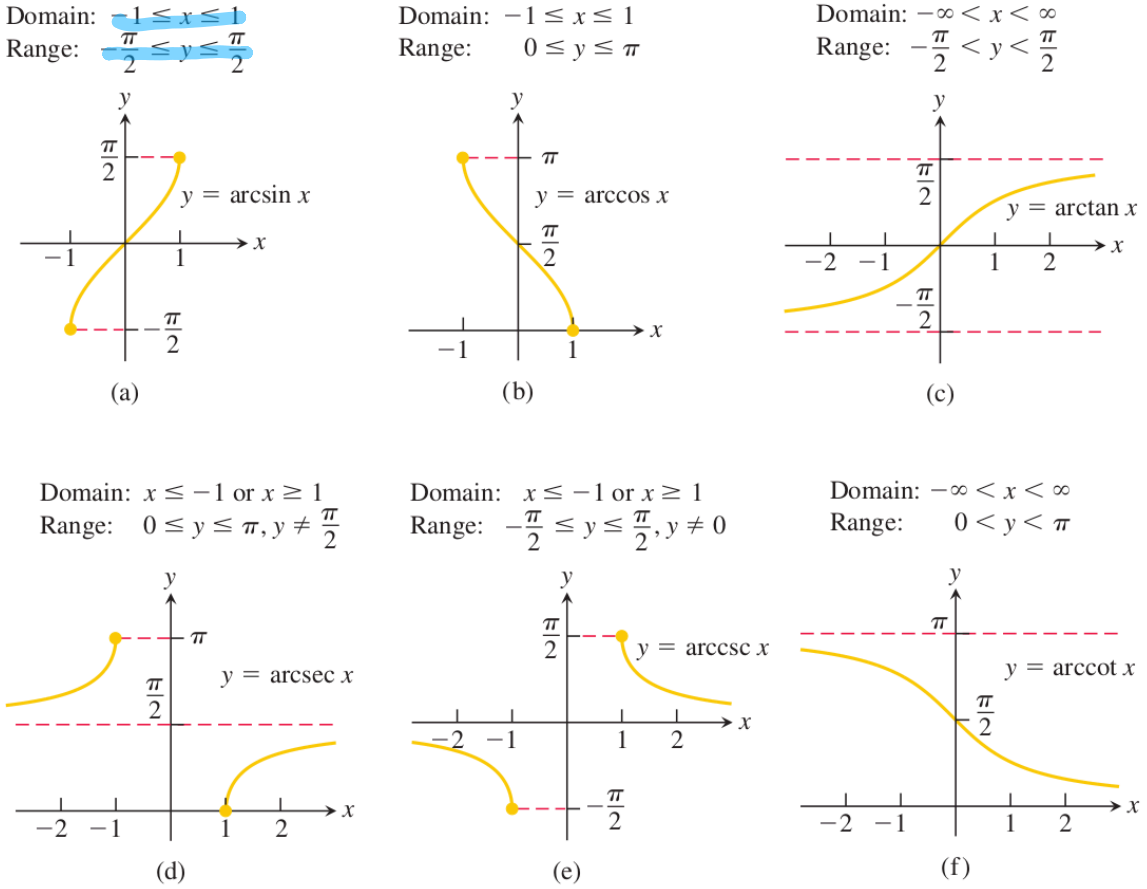


$y = \sec x$
Domain: $[0, \pi/2) \cup (\pi/2, \pi]$
Range: $(-\infty, -1] \cup [1, \infty)$



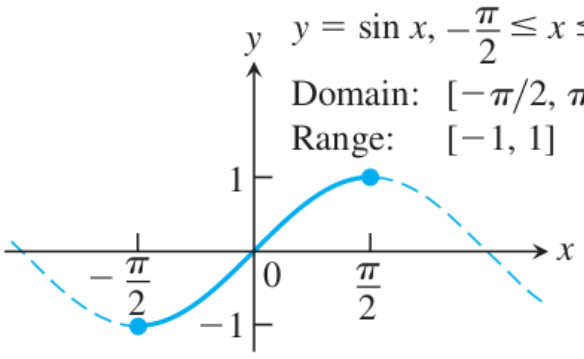
$y = \csc x$
Domain: $[-\pi/2, 0) \cup (0, \pi/2]$
Range: $(-\infty, -1] \cup [1, \infty)$

The graphs of inverse trigonometric functions.

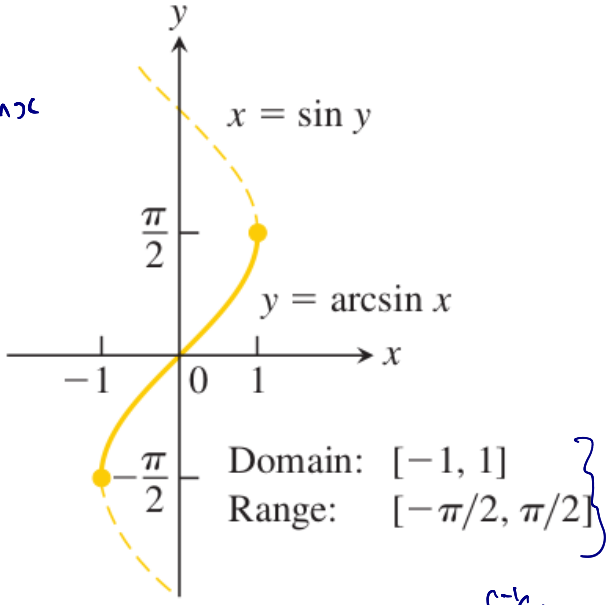


The Arcsine and Arccosine Functions

The function $\sin x$ is one-to-one on $[-\frac{\pi}{2}, \frac{\pi}{2}]$, so the inverse of sine exists. We let $y = \sin^{-1}(x)$ to be the value in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ for which $\sin y = x$.



$f(x) = \sin x$



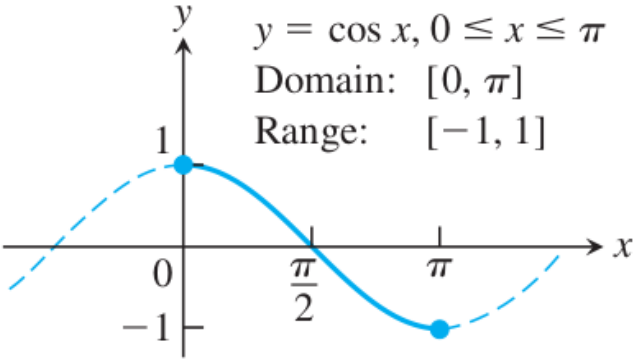
$f^{-1}(x) = \sin^{-1}(x)$

e.g. $\sin\left(\frac{\pi}{6}\right) = \frac{1}{2} \Rightarrow \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$

$\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2} \Rightarrow \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{3}$

Note: $\sin^{-1} x \neq \frac{1}{\sin x} = (\sin x)^{-1}$

The function $\cos x$ is one-to-one on $[0, \pi]$, so the inverse of cosine exists. We let $y = \cos^{-1}(x)$ to be the value in $[0, \pi]$ for which $\cos y = x$.

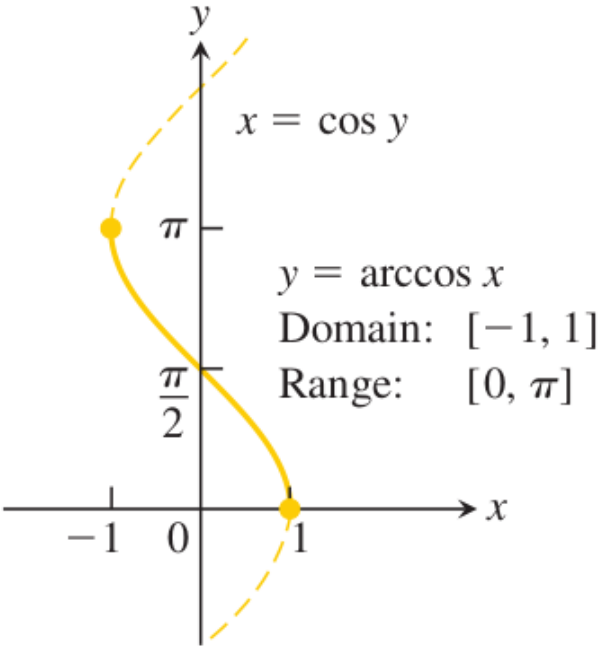


e.g. $\cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$

$\Rightarrow \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4}$

$\cos\left(\frac{\pi}{2}\right) = 0$

$\Rightarrow \cos^{-1}(0) = \frac{\pi}{2}$



Note: $\cos^{-1} x \neq \frac{1}{\cos x} = (\cos x)^{-1}$

Identities involving Arcsine and Arccosine

[$f(x)$ is odd $\Leftrightarrow f(-x) = -f(x)$].

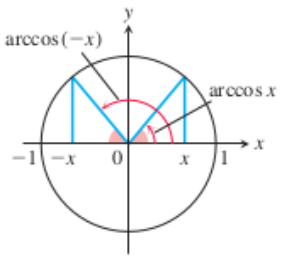


FIGURE 7.27 arccos x and arccos $(-x)$ are supplementary angles (so their sum is π).

$D_{\sin^{-1}(x)} = [-1, 1]$

$$\sin^{-1}(-x) = -\sin^{-1}(x)$$ ← Arcsin is odd.

$$\cos^{-1}(x) + \cos^{-1}(-x) = \pi$$

$$\sin^{-1}(x) + \cos^{-1}(x) = \frac{\pi}{2}$$ ($D_{\cos^{-1}} = D_{\sin^{-1}} = [-1, 1]$)

① $\sin^{-1}(x)$ is symmetric about the origin
⇒ it is an odd function.

② let $\cos^{-1}(x) = \theta$, then $\cos\theta = x$. ($\theta \in [0, \pi]$)

⇒ $\cos(\pi - \theta) = \cos\pi\cos\theta + \sin\pi\sin\theta$

$= -\cos\theta$

$= -x$

⇒ $\cos^{-1}(-x) = \pi - \theta$ ⇒ $\theta + \cos^{-1}(-x) = \pi$

⇒ $\cos^{-1}(x) + \cos^{-1}(-x) = \pi$

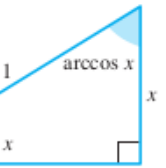
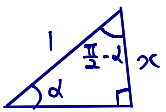


FIGURE 7.28 arcsin x and arccos x are complementary angles (so their sum is $\pi/2$).

③



let $\sin^{-1} x = \alpha$ ($x \geq 0$, $\alpha \in [0, \frac{\pi}{2}]$).

⇒ $\sin\alpha = x$

⇒ $\cos(\frac{\pi}{2} - \alpha) = x$

⇒ $\cos^{-1} x = \frac{\pi}{2} - \alpha$

⇒ $\alpha + \cos^{-1} x = \frac{\pi}{2}$ ⇒ $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$

($\because -x \geq 0$)

For $x \leq 0$

$\sin^{-1} x = -\sin^{-1}(-x)$

$= -[\frac{\pi}{2} - \cos^{-1}(-x)]$

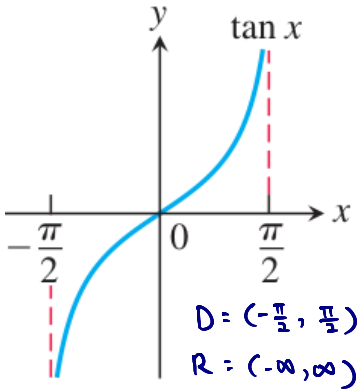
$= -\frac{\pi}{2} + \cos^{-1}(-x)$

$= -\frac{\pi}{2} + \pi - \cos^{-1} x$

⇒ $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$.

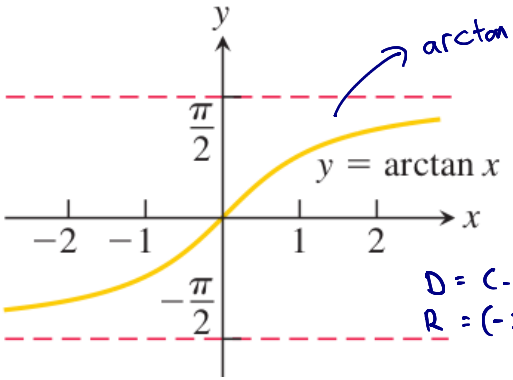
The Arctangent and Arccotangent Functions

The function $\tan x$ is one-to-one on $(-\frac{\pi}{2}, \frac{\pi}{2})$, so the inverse of tangent exists.



$D = (-\frac{\pi}{2}, \frac{\pi}{2})$

$R = (-\infty, \infty)$



arctan x is odd

$D = (-\infty, \infty)$

$R = (-\frac{\pi}{2}, \frac{\pi}{2})$

$\Rightarrow \arctan(-x) = -\arctan x$

The function $\cot x$ is one-to-one on $(0, \pi)$, so the inverse of cotangent exists.

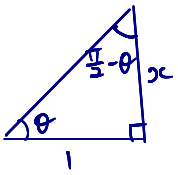
For $\cot x$, $D = (0, \pi)$ & $R = (-\infty, \infty)$

For $\operatorname{arccot} x$, $D = (-\infty, \infty)$ & $R = (0, \pi)$.

Identities involving Arctangent and Arccotangent

$$\tan^{-1}(-x) = -\tan^{-1}x \quad \text{(Arctangent is odd)}$$

$$\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2} \quad (D_{\tan^{-1}} = D_{\cot^{-1}} = \mathbb{R})$$



Let $\tan^{-1}x = \theta$, $x \geq 0$. $\Rightarrow \theta \in [0, \frac{\pi}{2})$

$\Rightarrow x = \tan\theta$

$\Rightarrow \cot(\frac{\pi}{2} - \theta) = x$

$\Rightarrow \cot^{-1}(x) = \frac{\pi}{2} - \theta$

$\Rightarrow \theta + \cot^{-1}(x) = \frac{\pi}{2}$

$\Rightarrow \underline{\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}}$

Remark: True for all $x \in \mathbb{R}$.

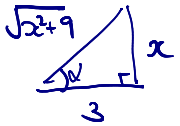
(Exercise: Verify for $x \leq 0$.)

Example:

a. Find $\sec\left(\tan^{-1}\frac{x}{3}\right)$.

let $\alpha = \tan^{-1}\frac{x}{3}$, then we want $\sec(\tan^{-1}\frac{x}{3}) = \sec\alpha$

$\Rightarrow \tan\alpha = \frac{x}{3}$



$\Rightarrow \sec\alpha = \frac{\sqrt{x^2+9}}{3}$

b. Find $\sin\left(\tan^{-1}\sqrt{x^2-2x}\right)$.

Ans: $\frac{\sqrt{x^2-2x}}{x-1}$

The inverses of secant and cosecant functions

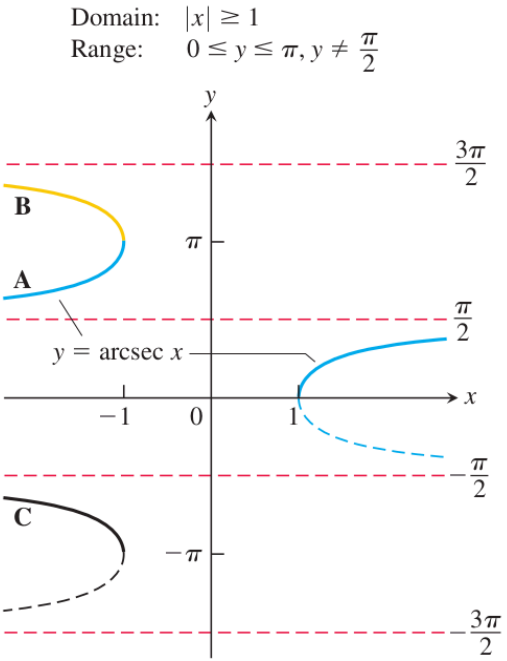


FIGURE 7.29 There are several logical choices for the left-hand branch of $y = \text{arcsec } x$. With choice **A**, $\text{arcsec } x = \arccos(1/x)$, a useful identity employed by many calculators.

By restricting the domain to $[0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$, the function sec x is 1-1. So $\sec^{-1}(x)$ exists for $|x| \geq 1$. Under such restrictions, we have the following identity.

$$\sec^{-1}(x) = \cos^{-1}(\tfrac{1}{x}) = \frac{\pi}{2} - \sin^{-1}(\tfrac{1}{x})$$

from earlier

let $\alpha = \sec^{-1} x$, then $x = \sec\alpha = \frac{1}{\cos\alpha}$

$\Rightarrow \cos\alpha = \frac{1}{x}$

$\Rightarrow \alpha = \cos^{-1}(\frac{1}{x})$

$\Rightarrow \sec^{-1} x = \cos^{-1}(\frac{1}{x})$

By restricting the domain to $[-\frac{\pi}{2}, 0) \cup (0, \frac{\pi}{2}]$, the function csc x is 1-1. So $\csc^{-1}(x)$ exists, for $|x| \geq 1$.

Identities Involving Inverse Trigonometric:

1. $\sin^{-1}(-x) = -\sin^{-1}x$	2. $\tan^{-1}(-x) = -\tan^{-1}x$
3. $\cos^{-1}x + \cos^{-1}(-x) = \pi$	4. $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$
5. $\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$	6. $\sec^{-1}x + \csc^{-1}x = \frac{\pi}{2}$
7. $\sec^{-1}x = \cos^{-1}\frac{1}{x}$	8. $\csc^{-1}x = \sin^{-1}\frac{1}{x}$
9. $\cot^{-1}x = \tan^{-1}\frac{1}{x}$	

Exercise : Verify identities 6, 8, 9.

The Derivative of $y = \sin^{-1} u$

Recall that $(f^{-1})'(x) = \dfrac{1}{f'(f^{-1}(x))}$

Let $f(x) = \sin x$, then $f^{-1}(x) = \sin^{-1} x$.

Recall that $\sin^2\theta + \cos^2\theta = 1$

$\Rightarrow \cos\theta = \pm\sqrt{1-\sin^2\theta}$

$f'(x) = \cos x$.

$$\Rightarrow \frac{d}{dx}\sin^{-1}x = \frac{1}{f'(f^{-1}(x))} = \frac{1}{f'(\sin^{-1}x)} = \frac{1}{\cos(\sin^{-1}x)} \quad (>0)$$

$$= \frac{1}{\pm\sqrt{1-\sin^2(\sin^{-1}x)}}$$

$$= \pm\frac{1}{\sqrt{1-x^2}} \qquad \left(\because \sin^2(\sin^{-1}x) = [\sin(\sin^{-1}x)]^2 = x^2\right)$$

We have $\sin^{-1}x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \Rightarrow \cos(\sin^{-1}x) > 0$

$$\Rightarrow \frac{d}{dx}\sin^{-1}x = \frac{1}{\sqrt{1-x^2}}, \quad |x|<1$$

Apply Chain Rule: $\dfrac{d}{dx}\sin^{-1}u = \dfrac{1}{\sqrt{1-u^2}}\dfrac{du}{dx}$, $|u|<1$, u is a differentiable function of x.

Example:

1. $\frac{d}{dx}(\sin^{-1}x^2) =$ $\frac{1}{\sqrt{1-u^2}}\frac{du}{dx}$ $= \frac{1}{\sqrt{1-x^4}} \cdot 2x$

$u = x^2$

$\frac{du}{dx} = 2x$

$= \frac{2x}{\sqrt{1-x^4}}$

2. $\frac{d}{dx}(\sin^{-1}\sqrt{1-x}) =$ $\frac{1}{\sqrt{1-u^2}}\frac{du}{dx}$

$u = \sqrt{1-x}$

$= \frac{1}{\sqrt{1-(1-x)}} \frac{-1}{2\sqrt{1-x}}$

$\frac{du}{dx} = \frac{-1}{2\sqrt{1-x}}$

$= -\frac{1}{2\sqrt{x(1-x)}}$

The Derivative of $y = \tan^{-1} u$

Let $f(x) = \tan x$, then $f^{-1}(x) = \tan^{-1} x$.

$f'(x) = \sec^2 x$

$$\Rightarrow \quad \frac{d}{dx} \tan^{-1} x = \frac{1}{f'(f^{-1}(x))}$$

$$= \frac{1}{\sec^2(\tan^{-1} x)}$$

Recall : $\sec^2\theta = 1 + \tan^2\theta$

$$= \frac{1}{1 + \tan^2(\tan^{-1} x)}$$

$$= \frac{1}{1 + x^2}$$

Chain Rule : $\frac{d}{dx} \tan^{-1} u = \frac{1}{1+u^2} \frac{du}{dx}$, u is a differentiable function of x.

Example:

1. $\frac{d}{dx}\tan^{-1}(1+x^2) =$ $\frac{1}{1+(1+x^2)^2}\frac{d}{dx}(1+x^2)$

$u = 1+x^2$

$= \frac{2x}{x^4+2x^2+2}$

2. $\frac{d}{dx}\ln(\tan^{-1}x) =$ $\frac{1}{\tan^{-1}x}\frac{d}{dx}\tan^{-1}x$

$= \frac{1}{\tan^{-1}x}\frac{1}{1+x^2}$

The Derivative of $y = \sec^{-1} u$

Instead of applying the inverse derivative theorem as before, we find the derivative of $y = \sec^{-1} x$, $|x| > 1$, using implicit differentiation and the Chain Rule:

$$y = \sec^{-1} x$$

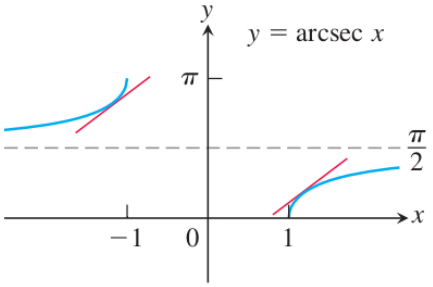


FIGURE 7.31 The slope of the curve $y = \text{arcsec}\, x$ is positive for both $x < -1$ and $x > 1$.

$\Rightarrow \sec y = x$

differentiate both sides wrt x $\Rightarrow \sec y \tan y \frac{dy}{dx} = 1$.

$\Rightarrow \frac{dy}{dx} = \frac{1}{\sec y \tan y} = \frac{1}{\pm x\sqrt{x^2-1}}$

$(\because \sec^2 y = 1 + \tan^2 y \Rightarrow \tan y = \pm\sqrt{\sec^2 y - 1})$

From the graph of $\sec^{-1} x$, we see that $\frac{dy}{dx} > 0$

So if $x > 1$, $\frac{dy}{dx} = \frac{1}{x\sqrt{x^2-1}}$

if $x < -1$, $\frac{dy}{dx} = -\frac{1}{x\sqrt{x^2-1}}$

$\Rightarrow \frac{dy}{dx} = \frac{1}{|x|\sqrt{x^2-1}}$

Chain Rule: $\frac{d}{dx}\sec^{-1} u = \frac{1}{|u|\sqrt{u^2-1}}\frac{du}{dx}$, $|u| > 1$, u diffble fn of x.

The Derivative of $y = \sec^{-1} u$

We can also get the derivative of $\sec^{-1}(x)$ using the identity $\sec^{-1}(x) = \cos^{-1}(\frac{1}{x}) = \frac{\pi}{2} - \sin^{-1}(\frac{1}{x})$.

Recall $\frac{d}{dx} \sin^{-1} u = \frac{1}{\sqrt{1-u^2}} \frac{du}{dx}$

$$\frac{d}{dx} \sec^{-1} x = \frac{d}{dx}\left(\frac{\pi}{2} - \sin^{-1}\left(\frac{1}{x}\right)\right)$$

$$= -\frac{d}{dx} \sin^{-1}\left(\frac{1}{x}\right), \quad u = \frac{1}{x}$$

$$= -\frac{1}{\sqrt{1-\left(\frac{1}{x}\right)^2}} \frac{d}{dx}\left(\frac{1}{x}\right)$$

$$= \frac{1}{x^2\sqrt{1-\frac{1}{x^2}}} = \frac{1}{\frac{x^2}{\sqrt{x^2}}\sqrt{x^2-1}}$$

$$= \frac{1}{\frac{x^2}{|x|}\sqrt{x^2-1}} = \frac{1}{|x|\sqrt{x^2-1}}$$

Derivative of the Other Three

The derivatives of the other three trigonometric functions are easier to derive using the following identities.

Inverse Function-Inverse Cofunction identities

$$\cos^{-1}x = \frac{\pi}{2} - \sin^{-1}x$$

$$\cot^{-1}x = \frac{\pi}{2} - \tan^{-1}x$$

$$\csc^{-1}x = \frac{\pi}{2} - \sec^{-1}x$$

- $\frac{d}{dx}\cos^{-1}x = -\frac{d}{dx}\sin^{-1}x = -\frac{1}{\sqrt{1-x^2}}$

- $\frac{d}{dx}\csc^{-1}x = \frac{d}{dx}\left(\frac{\pi}{2} - \sec^{-1}x\right) = -\frac{d}{dx}\sec^{-1}x = -\frac{1}{|x|\sqrt{x^2-1}}$

TABLE 7.3 Derivatives of the inverse trigonometric functions

1. $\dfrac{d(\arcsin u)}{dx} = \dfrac{1}{\sqrt{1-u^2}}\dfrac{du}{dx}, \quad |u| < 1$

2. $\dfrac{d(\arccos u)}{dx} = -\dfrac{1}{\sqrt{1-u^2}}\dfrac{du}{dx}, \quad |u| < 1$

3. $\dfrac{d(\arctan u)}{dx} = \dfrac{1}{1+u^2}\dfrac{du}{dx}$

4. $\dfrac{d(\operatorname{arccot} u)}{dx} = -\dfrac{1}{1+u^2}\dfrac{du}{dx}$

5. $\dfrac{d(\operatorname{arcsec} u)}{dx} = \dfrac{1}{|u|\sqrt{u^2-1}}\dfrac{du}{dx}, \quad |u| > 1$

6. $\dfrac{d(\operatorname{arccsc} u)}{dx} = -\dfrac{1}{|u|\sqrt{u^2-1}}\dfrac{du}{dx}, \quad |u| > 1$

Example: Find

1. $\frac{d}{dx}\sin^{-1}\sqrt{1-x},\ 0 < x < 1.$

2. $\frac{d}{dx}\tan^{-1}(1+x^2)$

3. $\frac{d}{dx}\sec^{-1}(\frac{e^{2x}}{2}),\ x > \frac{1}{2}.$

4. $\frac{d}{dx}\sec^{-1}(\frac{1}{x}),\ 0 < x < 1.$

5. $\frac{d}{dx}\csc^{-1}(x^3)$

6. $\frac{d}{dx}\cos^{-1}(\frac{x-1}{x+1})$

$$\frac{1}{|u|\sqrt{u^2-1}}\frac{du}{dx} = \frac{1}{|\frac{e^{2x}}{2}|\sqrt{\frac{e^{4x}}{4}-1}}\frac{d}{dx}\left(\frac{e^{2x}}{2}\right)$$

$$= \frac{1}{\frac{e^{2x}}{2}\sqrt{\frac{e^{4x}}{4}-1}}\,e^{2x}$$

$$= \frac{4}{\sqrt{e^{4x}-4}}$$

$u = \frac{e^{2x}}{2}$

Integrals evaluated with inverse trigonometric functions

Recall that $\frac{d}{dx}\sin^{-1}x = \frac{1}{\sqrt{1-x^2}}$

Let u be a diff. function of x, and we let a be a constant such that $u^2 < a^2$, $a > 0$.

Then $\frac{d}{dx}\sin^{-1}\left(\frac{u}{a}\right) = \frac{1}{\sqrt{1-\frac{u^2}{a^2}}}\frac{d}{dx}\left(\frac{u}{a}\right)$

$= \frac{1}{a\sqrt{1-\frac{u^2}{a^2}}}\frac{du}{dx} = \frac{1}{\sqrt{a^2-u^2}}\frac{du}{dx}$

$\Rightarrow$ (integrate both sides) $\int \frac{1}{\sqrt{a^2-u^2}}\,du = \sin^{-1}\left(\frac{u}{a}\right) + C$

TABLE 7.4 Integrals evaluated with inverse trigonometric functions

The following formulas hold for any constant $a > 0$.

1. $\displaystyle\int \frac{du}{\sqrt{a^2 - u^2}} = \sin^{-1}\left(\frac{u}{a}\right) + C$ (Valid for $u^2 < a^2$)

2. $\displaystyle\int \frac{du}{a^2 + u^2} = \frac{1}{a}\tan^{-1}\left(\frac{u}{a}\right) + C$ (Valid for all u)

3. $\displaystyle\int \frac{du}{u\sqrt{u^2 - a^2}} = \frac{1}{a}\sec^{-1}\left|\frac{u}{a}\right| + C$ (Valid for $|u| > a > 0$)

$$\frac{d}{dx}\tan^{-1}\left(\frac{u}{a}\right) = \frac{1}{1+\frac{u^2}{a^2}}\cdot\frac{d}{dx}\left(\frac{u}{a}\right) = \frac{1}{a+\frac{u^2}{a}}\frac{du}{dx}$$

$$= \frac{a}{a^2+u^2}\frac{du}{dx}$$

$$\Rightarrow \int \frac{a}{a^2+u^2}\,du = \tan^{-1}\left(\frac{u}{a}\right) + C$$

Example:

1. $\int \frac{dx}{\sqrt{3-4x^2}} = \int \frac{1}{\sqrt{a^2-u^2}} \frac{du}{2}$

$a = \sqrt{3}$ $= \frac{1}{2}\int \frac{1}{\sqrt{a^2-u^2}}\,du$

$u = 2x$

$\frac{du}{dx} = 2$ $= \frac{1}{2} \sin^{-1}\left(\frac{u}{a}\right) + C$

$= \frac{1}{2} \sin^{-1}\left(\frac{2x}{\sqrt{3}}\right) + C$

2. $\int \frac{dx}{\sqrt{4x-x^2}} = \int \frac{1}{\sqrt{4-(x-2)^2}}\,dx$

Complete the square

$4x - x^2 = -(x^2 - 4x)$

$= -[(x-2)^2 - 4]$

$= 4 - (x-2)^2$

$= \int \frac{1}{\sqrt{a^2-u^2}}\,du$

$= \sin^{-1}\left(\frac{x-2}{2}\right) + C$

$a = 2$, $u = x - 2$

3. $\int \frac{dx}{4x^2+4x+2} = \int \frac{1}{(2x+1)^2+1}\,dx$

$a = 1$

$u : 2x+1$

$du : 2\,dx$

$= \frac{1}{2}\int \frac{1}{a^2+u^2}\,du$

$= \frac{1}{2}\,\frac{1}{a}\tan^{-1}\left(\frac{u}{a}\right) + C$

$= \underline{\underline{\frac{1}{2}\tan^{-1}(2x+1) + C}}$

4. $\int \frac{dx}{\sqrt{e^{2x}-6}} = \int \frac{du}{u\sqrt{u^2-a^2}}$

$u = e^x \qquad a = \sqrt{6}$

$du = e^x\,dx$

$\Rightarrow dx = \frac{du}{e^x}$

$= \frac{du}{u}$

$= \frac{1}{a}\sec^{-1}\left(\left|\frac{u}{a}\right|\right) + C$

$= \underline{\underline{\frac{1}{\sqrt{6}}\sec^{-1}\left(\frac{e^x}{\sqrt{6}}\right) + C}}$